Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **Rockers**
b. Team leader name: **Hafiz Rizwan Umar**
c. Problem Statement: Teachers in emerging markets lose 3+ hours per assignment cycle to manual admin and paper reports, while integrity tools stay unaffordable.

Challenge: Sustainability & Social Impact | **Live prototype: https://www.inklesslms.com**

Brief about the idea

Inkless replaces five disconnected EdTech tools with one AI-powered LMS.

Problem. A teacher with 200 students downloads every ZIP, extracts it, opens files in separate apps, then grades. 43% of students use AI to cheat, and most teachers can only spot-check.

Idea. One browser-based LMS where submissions open instantly in tabs, every submission is auto-checked for plagiarism, AI authorship and collusion, and AI generates KPI-weighted quizzes and accreditation reports.

Why it matters. Zero paper, zero setup, and a free tier give universities in developing countries integrity tools that only wealthy institutions could afford.

5 → 1

tools replaced by one platform

3+ hrs → 8 sec

to open and review a 200-student ZIP cycle

150M+

Classroom users without these features

Proof, not a pitch: live at **inklesslms.com** and deployed at Riphah International University, Pakistan.

Recognition: Shortlisted among Asia's Top 20 Best Startups for Sustainability • 3× startup competition winner (selected from 1,400+ startups, Pakistan National Incubation Center) • Peer-reviewed research on Google Classroom gaps.

Opportunities

How is it different?

Google Classroom: no plagiarism detection.

Turnitin alone: manual and disconnected.

Quiz tools: no KPI weighting, no anti-cheat.

Inkless joins all of it in one workflow, with integrity checks that run on every submission automatically.

How will it solve it?

- ZIP files auto-extract into browser tabs: no downloads.
- Turnitin API checks similarity, AI authorship and inter-student collusion.
- AI builds quizzes with per-question KPI weights and live anti-cheat.
- Reports and CLO mapping generate in one click.

USP of the solution

- Cohort-level similarity engine (proprietary).
- Only LMS combining a smart submission viewer, auto-integrity, KPI-weighted AI quizzes and accreditation reports.
- Live, in production, mobile-responsive.
- Freemium: $0 entry, $8/teacher/month Pro.

Market opportunity: $340B global EdTech market • 50,000+ universities in Asia + MENA • $12B LMS + integrity serviceable market. Go-to-market is bottom-up: one free teacher adopts, the department follows.

List of features offered by the solution

Smart ZIP Viewer

All submitted files open in browser tabs. No downloads, works on mobile.

Live Code Review

Built-in code viewer so teachers grade inline, in one place.

Auto Integrity

Turnitin API runs on every submission: similarity, AI authorship, collusion.

Cohort Similarity Engine

Proprietary inter-student similarity detection across a whole class.

AI Quiz Engine

Generates targeted quizzes from course material, with KPI weights per question.

Anti-Cheat Monitoring

Live monitoring during quizzes and assessments.

Auto-CLO Mapping

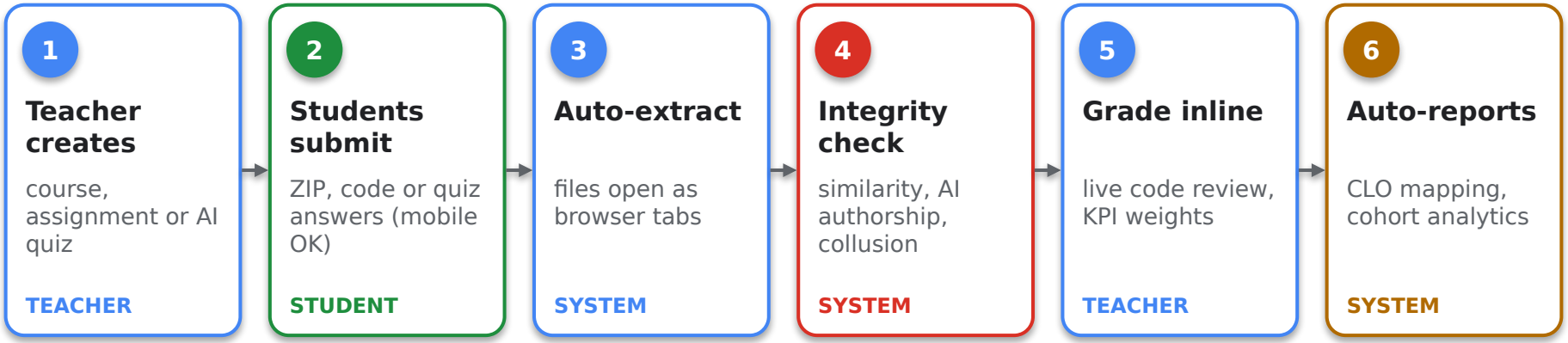
Maps student progress to Course Learning Outcomes for accreditation.

Instant Reports

Cohort analytics and KPI attainment, downloadable in one click.

Plus: freemium access (Free / Pro / Enterprise), mobile-responsive UI, zero paper, zero IT setup.

Process flow diagram or Use-case diagram



BEFORE

Download ZIP → extract → open in app → grade → repeat 199 times. Export to Turnitin, wait, cross-reference. Build quizzes and reports by hand.

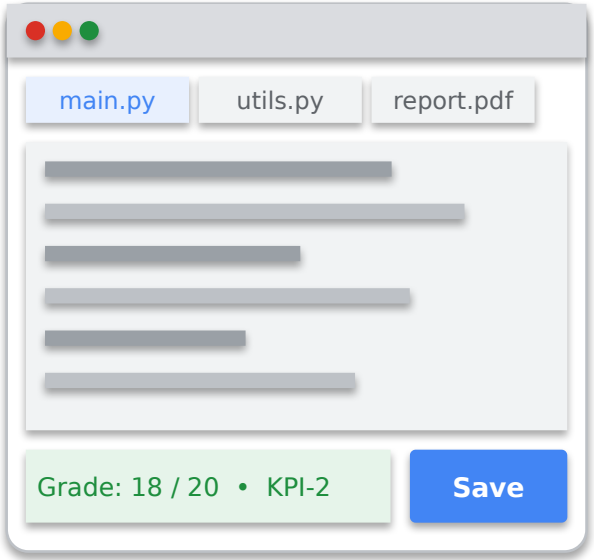
3+ hours of admin per cycle

WITH INKLESS

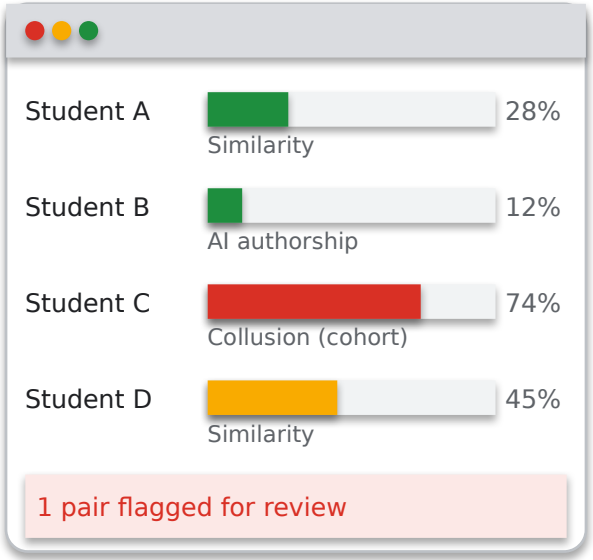
Submissions open instantly, integrity runs by itself, AI writes the quiz, and the report is generated automatically. Zero paper.

About 8 seconds, not 3 hours

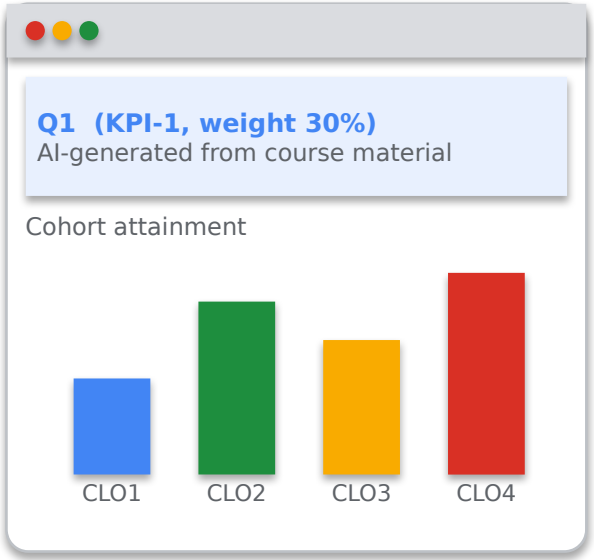
Wireframes/Mock diagrams of the proposed solution (optional)



Smart Submission Viewer



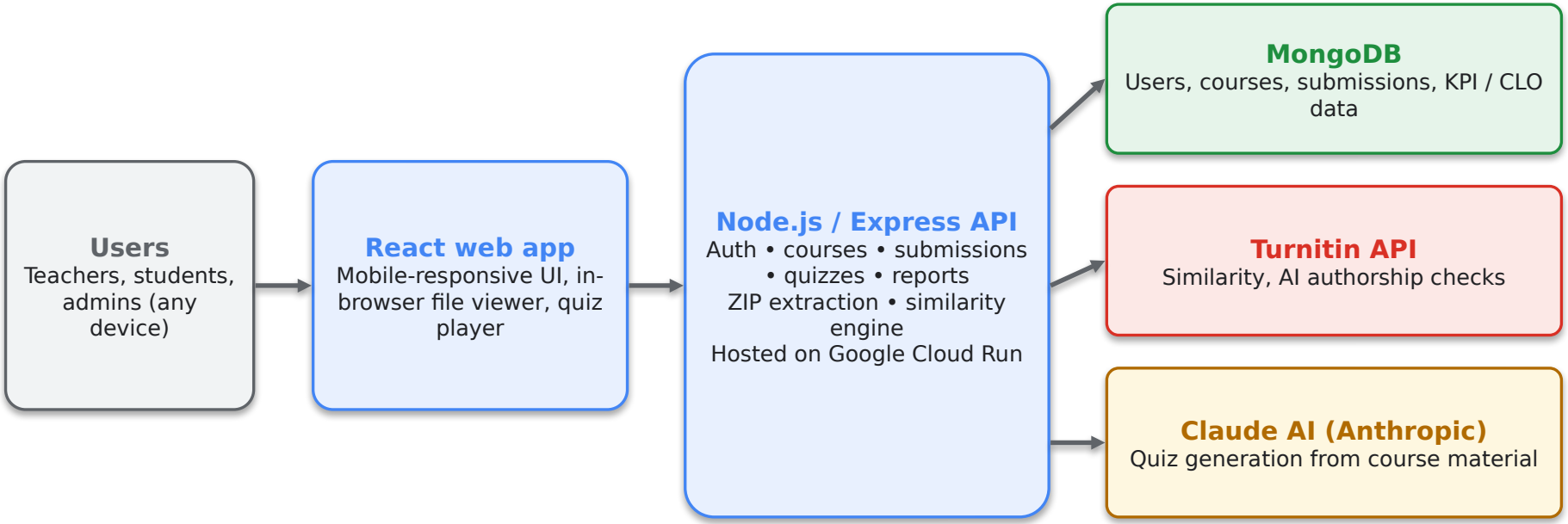
Auto Integrity Report



AI Quiz + KPI Dashboard

Illustrative wireframes of the live prototype at inklesslms.com (full screenshots on slide 10).

Architecture diagram of the proposed solution



Cloud-native SaaS: stateless API, managed hosting, zero paper, scales per institution.

Technologies to be used in the solution

Frontend

- React (MERN stack)
- Mobile-responsive design
- In-browser ZIP / code viewer

Backend & data

- Node.js + Express
- MongoDB
- Production-grade REST API

AI & integrity

- Anthropic Claude: quiz generation
- Turnitin API: plagiarism + AI authorship
- Proprietary cohort similarity engine

Cloud

- Google Cloud Run deployment
- Live at inklesslms.com
- Freemium, usage-based scaling

Why this stack: a single JavaScript codebase keeps the team lean, managed cloud hosting removes the need for on-campus servers, and AI plus Turnitin run behind the scenes so teachers take zero extra steps.

Estimated implementation cost (optional)

Use of the $150K pre-seed plan



- 40% Product dev (mobile, analytics)
- 30% Sales & marketing
- 20% Team (1 sales hire)
- 10% Legal & ops (IP, entity)

Pricing that keeps access open

Free — **$0**

Up to 3 courses, basic LMS

Pro — **$8 / teacher / month**

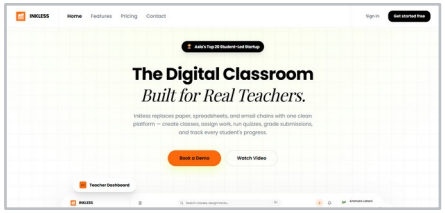
Turnitin integration, AI quizzes, KPI reports, anti-cheat

Enterprise — **$2K–$10K / year**

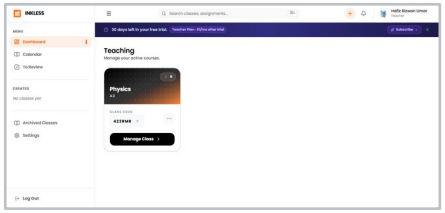
Full platform, white-label, API access, support

Cloud-native and serverless-friendly: running cost scales with usage, so a free teacher costs almost nothing to serve.

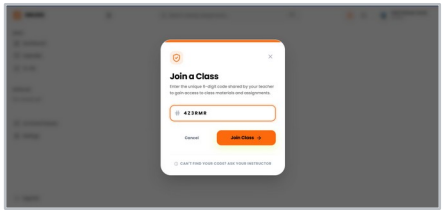
Snapshots of the prototype



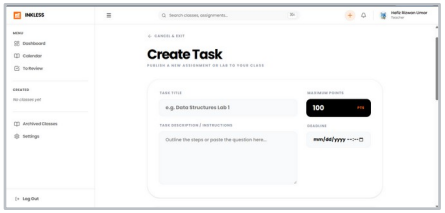
Landing page



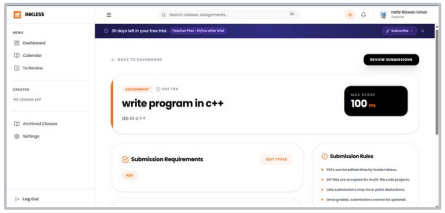
Teacher dashboard



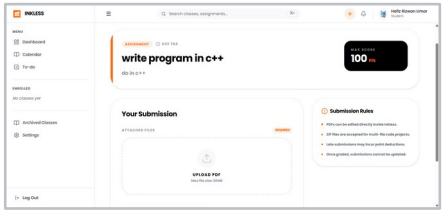
Student: join a class by code



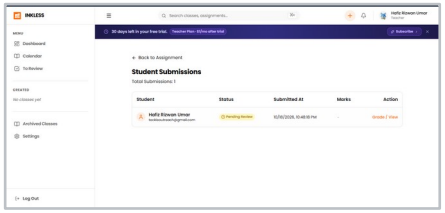
Teacher: create task



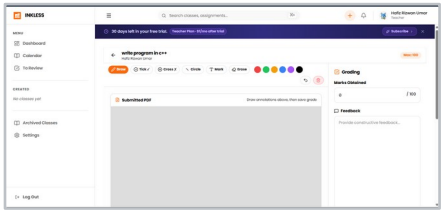
Assignment view



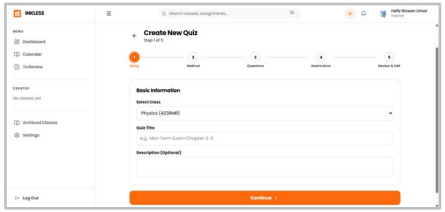
Student submission upload



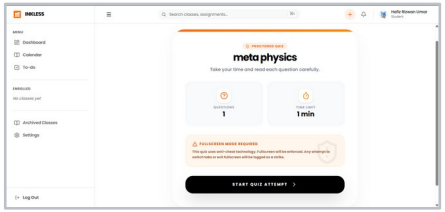
Submissions tracker



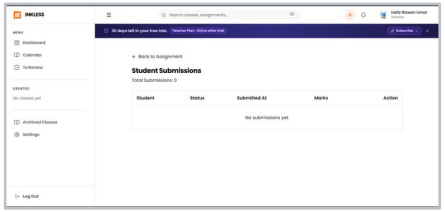
PDF grading + annotations



Quiz builder (5-step wizard)



Proctored quiz (anti-cheat)



Submissions (empty)

Live prototype

inklesslms.com

Real screens from the deployed app

Live prototype: https://www.inklesslms.com

Prototype Performance report/Benchmarking

~8 sec
vs 3+ hrs to open a 200-student cycle

2–3 hrs
saved per cycle at Riphah University

100%
of submissions auto-checked (vs spot-checks)

5 → 1
tools replaced

Capability	Google Classroom	Turnitin alone	Inkless
In-browser ZIP / code viewer	No	No	**Yes**
Auto plagiarism + AI-authorship check	No	Manual	**Automatic**
Inter-student (cohort) similarity	No	Limited	**Yes**
KPI-weighted AI quiz generation	No	No	**Yes**
Anti-cheat quiz monitoring	No	No	**Yes**
Auto-CLO mapping + accreditation reports	No	No	**Yes**

Sources: Inkless deployment at Riphah International University; competitor gaps per the Inkless research paper on Google Classroom (ResearchGate / Academia.edu); 43% AI-cheating figure from Cotton et al. (2024).

Sustainability & Social Impact (our challenge track)

Environmental

- Cloud-native SaaS, zero paper.
- Eliminates printing of assignments and reports across thousands of students.
- No on-campus servers to power or maintain.

Social

- Integrity tools once affordable only to wealthy institutions, now open to developing-country universities.
- Mobile-first access for students on phones.
- Fair, consistent checking of every submission.

Economic

- Freemium: access regardless of institutional budget.
- Gives teachers back 2–3 hours per cycle.
- Creates local EdTech jobs in Pakistan.

Recognised: Shortlisted among Asia's Top 20 Best Startups for Sustainability. Live at Riphah International University, 3× startup competition winner, peer-reviewed research on education-technology gaps.

Additional Details/Future Development (if any)

0–6 months

- Onboard 5 universities (Pakistan + Middle East)
- Launch Inkless Pro at $8/teacher/month
- Apply to the Google for Education programme
- File patent for cohort similarity detection

6–18 months

- 50+ universities across South Asia and MENA
- Launch Inkless Enterprise
- Turnitin preferred-partner agreement
- iOS + Android mobile app

18–36 months

- 500+ universities, international expansion
- Curriculum intelligence analytics product
- Ministry-level white-label deals
- Target: $5M ARR in year 3

Already achieved: platform live at inklesslms.com • adopted by Riphah International University • research paper published • 3× startup competition winner • Asia Top 20 sustainability shortlist • pipeline of 3 universities in conversation, outreach to 8 more across UAE, Malaysia and the UK.

Provide links to your:

GitHub Public Repository

https://github.com/HafizRizwanUmar/inklesslms

Final Product Link

https://www.inklesslms.com

Challenge: **Sustainability & Social Impact** | Contact: hafiz@minderfly.com

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

